	7.12 CHARACTERISATION
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Characterisation

Objectives

To assign a property value to a reference material using scientifically valid, traceable, and statistically sound techniques.

1. Scope

This procedure describes how to determine the certified values of reference materials (RMs/CRMs) through characterisation in compliance with ISO 17034:2016 Clause 7.12.

2. Reference Standards

- ISO 17034:2016 — General requirements for the competence of RMPs
- ISO/IEC 17025 — General requirements for the competence of testing and calibration laboratories
- ISO 33405:2024— Reference materials – Guidance for characterisation and assessment of homogeneity and stability
- Specific test method standards (e.g. ISO 1171, ISO 562, ISO 589, etc.)

3. Responsibilities

- Analyst: Conducts characterisation measurements
- Laboratory Manager: Process data, calculate assigned values, reviews and approves characterisation reports
- Operations Manager: Reviews and approves all procedures, technical forms.

4. Characterisation Strategy

The lab will be using the following approaches:

- One laboratory using validated methods

- Multiple laboratories using the same or different validated methods
- Combination of methods or historical data, where justified.

5. Procedure step

5.1. Define the Measurand

Clearly state the property (e.g., Ash content in % on dry basis) and units.

5.2. Select the Measurement Method

- Use methods validated per ISO/IEC 17025 or ISO 5725.
- The method must be appropriate for the measurand and matrix.
- Document all method parameters (temperature, mass, sample prep, etc.).

5.3. Ensure Metrological Traceability

- Use instruments calibrated with standards traceable to the SI units.
- Use CRM reference materials if applicable (e.g., for calorific value).

5.4. Perform the Measurements

- Use at least **2 replicate measurements per unit** to ensure repeatability.
- If multiple units are used, select them randomly.
- If multiple labs are involved, standardise procedures and sample prep.

5.5. Control Environmental Conditions

Ensure environmental conditions (e.g., temperature, humidity) are suitable for accurate and stable measurements.

5.6. Evaluate and Document Raw Data

- Record all instrument settings, analyst IDs, and time/date.
- Secure all records in a traceable, audit-proof system.

5.7. Statistical Treatment

- Use means or robust statistical estimators (e.g., robust mean, median).
- Identify and eliminate outliers using ISO 5725, Dixon's Q, or Grubbs' test or any other sound statistical means.
- Calculate the standard deviation of the mean or robust uncertainty.

5.8. Assess Method Uncertainty

- Include method repeatability, bias, calibration, and sample preparation effects.
- Uncertainty should be documented in an uncertainty budget.

5.9. Assign Property Values

- The assigned value is the best estimate of the true value based on valid statistical analysis.
- Include expanded uncertainty ($U = k \times u$), coverage factor (usually k=2 for 95% confidence).

5.10. Peer Review / Technical Review

- Independent review by technical staff not involved in measurement.
- Check method validation, traceability, uncertainty, and data integrity.

6. Documentation and Reporting

The following should be included in the characterisation report:

- RM Name, ID, Batch
- Measurand and method used
- Measurement results
- Assigned value and uncertainty
- Traceability chain
- Analysts, labs involved
- Statistical treatment

The records should be archived for a period of 5 years.

7. Controls and Records

Record	Description
Raw data sheets	Original measurement results
Calibration certificates	For all measurement instruments used
Uncertainty budget	Used to estimate uncertainty
Statistical reports	All analysis used to determine certified values
Characterisation reports	Final signed report for each batch